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Modeling a Multitarget Attacker–Defender Game with Budget Constraints

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Abstract. Though the choices of terrorists' attack targets are vast, their resources are limited. In this paper, a game-theoretical model is proposed to study both the defender's (government) and the attacker's (terrorist) expenditures among multiple targets under budget constraints to guide investment in defense. We study how the defender's and the attacker's equilibrium allocations depend on the budget constraints, target valuations, cost effectiveness of their investments, and inherent defense levels of targets in both sequential-move and simultaneous-move games. The equilibrium solutions are provided using the Karush–Kuhn–Tucker conditions. At the subgame-perfect Nash equilibrium, the defender's total marginal effects are the same among targets. Moreover, the defender's total marginal effects can be decomposed into direct and indirect effects. We also use the multiple-infrastructure and multiple-urban-area data sets to demonstrate the model results. The regression analysis shows that both the attacker's and the defender's equilibrium investments increase with their own target valuations, because a higher valuation for themselves means a higher attractiveness. Interestingly, the attacker's equilibrium investment is negatively correlated with the defender's target valuations, since a higher defender's valuation would make it more difficult for the attacker to successfully attack the target. By contrast, the defender's equilibrium investment is positively correlated with the attacker's target valuations, as a higher attacker's valuations would increase the urgency for the defender to protect the target. To show the utility of the new model, we compare the results of this model and the model in which the defender assumes that only a single target will be attacked when there could actually be multiple targets attacked. Our results show that the defender will suffer higher expected losses if she assumes that the attacker will attack only one target. The analysis of the attacker's and the defender's budget constraints show that (a) the higher the budget the defender has, the less likely it is that her most valuable target will be attacked; (b) a higher proportion of defense resources should be allocated to the most valuable target if the defender's budget is low; and (c) the attacker is less concentrated on attacking the most valuable target and spreads the resources to attack more targets as his budget increases.

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Keywords: terrorism and counter terrorism • Karush–Kuhn–Tucker condition • budget constraint • sequential game • simultaneous game • subgame perfect Nash equilibrium

1. Introduction

Protecting the nation from terrorism is an important but challenging problem faced by the government. The U.S. Department of Homeland Security (DHS) has spent hundreds of billions of dollars on counterterrorism. From fiscal year (FY) 2003 to fiscal year 2017,

the DHS annual budget increased from \$37.7 billion to \$64.8 billion (U.S. Department of Homeland Security 2017). Regardless of budget, how to best use defensive resources against terrorism remains a challenge, especially when facing adaptive adversaries. In 2009, the DHS launched an efficiency review program to

track the expenditure of the department operation and to look for an optimal way to spend limited taxpayer dollars effectively and efficiently (U.S. Department of Homeland Security 2016).

Many researchers have studied the counterterrorism problem either without considering the budget constraint problem or exclusively considering the government's budget constraint problem in the literature (Hausken and Levitin 2009, Hausken and Zhuang 2013, Levitin and Hausken 2009a, Shan and Zhuang 2013a, Wang and Bier 2015). However, as pointed out by Shughart and William (2011), a budget limitation is faced by not only the government but also the terrorist. With their financial capabilities, terrorist groups could invest their resources in organizational capabilities (e.g., leadership, publicity, recruitment) or operational capabilities (e.g., weapons, trainings, logistics), long-term investment (to establish capabilities) or short-term expenses (to execute attacks). Moreover, the effectiveness of investments is very different when the attacker is planning a series of improvised explosive device (IED) attacks on multiple homogenous targets (such as subway stations in one or more cities), rather than when the attacker is preparing for a spectrum of different attack types (hijacking, IED, or cyber attacks). Thus, to better understand the terrorists' activities and obtain a better defensive strategy, considering their financial activities and situation is important. Farrow (2007) proposed optimization models to study the strategies of both the attacker and the defender with the consideration of their budget constraints separately, but failed to link them together. In this paper, we study the budget constraints for both the attacker and the defender in one model.

Because of asymmetric information between the government and the terrorist (Sandler 2005), obtaining the terrorist's financial information remains a challenging problem. The government studied terrorist financing after the September 11, 2001, attack (Donohue 2006) to better defend against the terrorists' activities, which are supported by the terrorist's resources. The government then launched a terrorist finance tracking program to identify and track the transactions between terrorist cells, making it possible to estimate the terrorist's budget since 2001. For example, the Central Intelligence Agency estimated that Al-Qaeda spent approximately \$30 million per year prior to September 11, 2001, to

support the Taliban for operative training camps in Afghanistan, creating terrorist networks and alliances, and supporting the jihadists and their families (U.S. Department of the Treasury 2011). With the terrorist finance tracking program, the assessment of the terrorists' financial information becomes possible.

Levitin and Hausken (2009a) and Hausken and Levitin (2009) study the optimal resource distribution between deploying false targets and protecting genuine elements, given constant attack and defense budgets under the assumption that the attacker would only attack one target at a time. Many counterterrorism models in the literature also assume that the attacker's effort is in a discrete manner (attack or not attack) and the attacker will attack only one target at a time (Bier et al. 2008; Nikoofal and Zhuang 2012; Shan and Zhuang 2013b, c). There are also researchers studying continuous attacker efforts among multiple attacks, for example, Zhuang and Bier (2007), Hausken and Levitin (2010), and Wang and Bier (2015), etc. Levitin and Hausken (2009b) study redundancy versus protection in defending parallel systems, considering both the defender's and the attacker's budget constraints and multiple-target (MT) attacks, but they treat all the targets as identical. Many recent events (Enders and Sandler 2006, Hartwell and Barkley 2011) show that terrorists do attack multiple targets at the same time, which causes more damage and injuries to the public. For example, on September 11, 2001, four targets were attacked in the United States through a series of hijacked airplanes causing more than \$90 billion in economic losses and around 3,000 casualties; on July 7, 2005, four transportation targets were attacked in London, United Kingdom, causing 52 fatalities and over 700 injuries; on December 8, 2009, a police patrol and government buildings were simultaneously attacked by five bombings in Baghdad, Iraq, and at least 127 people were killed and 400 people were wounded (Wikipedia 2014). It is realistic to assume that multiple targets could be attacked at one time. Different targets have different value for both the attacker and the defender. For example, New York City would be valued much higher than a desert area. In this paper, we model the attack and defense efforts in a continuous manner with budget constraints, and we also allow the attacker to launch multiple attacks to different multiple targets.

To capture the essential relationships among the probability of a successful attack, defense and attack efforts, and the inherent defense levels, two forms of contest success functions (CSFs) among the players in the field of rent seeking, tournaments, and conflict (RMS 2011) are introduced by Hirshleifer (1989) and Skaperdas (1996). One is the exponential form (Shan and Zhuang 2013a, Wang and Bier 2011) and the other is the ratio form (Hausken and Zhuang 2013, Hirshleifer 1989, Zhuang and Bier 2007). The CSFs are assumed to be twice differentiable, continuous, and with diminishing marginal returns with respect to both the defense and attack efforts. Zhuang and Bier (2007), Levitin and Hausken (2010), and Wang and Bier (2015) allowed for continuous attacker efforts among multiple attacks, but did not consider budget constraints. In this paper, we model the relationships among the probability of successful attacks, defense and attack efforts, and the inherent defense levels by using the CSFs with the general CSFs feature, with consideration of budget constraints. We do not consider uncertainty of defender or attacker in the game, bounded rationality of players, or interdependencies between different targets as studied by Wang and Bier (2012), Shan and Zhuang (2013c), and Bier et al. (2007).

To study the conflict between the defender and attacker, Daniel et al. (2003) and Sandler and Siqueira (2009) show that game theory is an appropriate tool for examining the strategic interaction between the defender and the attacker. Game theory is widely used in studying multiplayer resource allocation problems in various fields. Based on whether the attacker observes the defender's investment decision before he makes the attack decision, both sequential-move game (Azaiez and Bier 2007, Wang and Bier 2011, Shan and Zhuang 2013c) and simultaneous-move games (Heal and Kunreuther 2007, Hausken et al. 2008, Dighe et al. 2009, Sandler and Siqueira 2009) have been studied.

The resource restriction problem is faced by not only the government but also the terrorist. Since studies on the counterterrorism problem with multiple targets and with resource restrictions for both the government and the terrorists were missing, we fill the gap by first addressing such a problem with the constraint of both the terrorist's and the government's limited resources. This paper studies an attacker–defender game model by considering the budget constraints for both players.

The defender and the attacker are assumed to allocate a continuous level of resources among multiple targets. The rest of this paper is organized as follows: Section 2 explains the notations and assumptions, proposes the optimization models for both the attacker and the defender, and studies their best response functions. Section 3 provides the equilibrium strategies in sequential-move and simultaneous-move games. Section 4 numerically illustrates the equilibrium solutions by using a multiple-infrastructure protection game and a multiple-urban-area protection game as two examples, and compares the results of the model with the scenario where the defender assumes that the attacker would only attack a single target (ST). Section 5 summarizes this paper and provides some future research directions. The appendix provides the proofs to all propositions.

2. Notation, Assumptions, and Optimization Problems

2.1. Notations

The notations used throughout this paper are defined in Table 1. There are $n \geq 1$ targets considered in this paper, which the defender needs to defend and the attacker could attack by allocating their resources $G = [G_1, G_2, \dots, G_n]$ and $T = [T_1, T_2, \dots, T_n]$, respectively. Both the resource allocation strategies of the defender and the attacker are subject to their budget constraints, which are $\mathcal{G} = \sum_i^n G_i$ and $\mathcal{T} = \sum_i^n T_i$, respectively. The defender and attacker are allowed to evaluate the target's value differently, as $D_i \geq 0$ and $B_i \geq 0$, respectively. The coefficients $\alpha_i \geq 0$ and $\beta_i \geq 0$ are used to model the effectiveness of the defender's and the attacker's investment, respectively. For each target, the existing defense system level is modeled by the inherent defense level, A_i , $i = 1, 2, \dots, n$. As we need to solve the constraint optimization problems, we denote the Lagrangian multipliers in solving a simultaneous-move game as λ_T^C , λ_G^C , $\mu_{T_i}^C$, and $\mu_{G_i}^C$, and those in solving a sequential-move game as λ_T^S , λ_G^S , $\mu_{T_i}^S$, and $\mu_{G_i}^S$, which are also called the shadow prices. In constrained optimization, the shadow price is the instantaneous change per unit of the constraint in the optimal objective solution obtained by relaxing the constraint (Bowles 2009, Gans et al. 2011). The equilibrium solutions for a simultaneous-move game and for a sequential-move game are denoted as (T^C, G^C) and (T^S, G^S) , respectively.

2.2. Assumptions

Following Azaiez and Bier (2007), Wang and Bier (2011), and Shan and Zhuang (2013c), both the attacker and the defender are assumed to be rational and strategic in this paper. In this paper, we model the interaction between the attacker and the defender by both sequential-move game and simultaneous-move game. Each player is assumed to know the complete information of the other, including the total budget and target valuation of the other player. The attacker and defender are allowed to have different target valuations. In the sequential-move game, the defender is assumed to move first. After observing the defender's investment decision $G \equiv (G_1, G_2, \dots, G_n)$, the attacker allocates his attack resource $\mathcal{T}$ among the n targets $T \equiv (T_1, T_2, \dots, T_n)$, subject to a budget constraint $\sum_{i=1}^n T_i = \mathcal{T}$. In the simultaneous-move game, the attacker and the defender are assumed to make

decisions simultaneously without knowing the other's decision.

Both the attacker's and defender's strategies T_i and G_i impact the probability of a successful attack $P_i(T_i, G_i)$ on the i th target. The probability function is assumed to be continuous and twice differentiable. It is also assumed to have the following properties (Hausken and Zhuang 2013, Hirshleifer 1989, Zhuang and Bier 2007):

$$\begin{aligned} \frac{\partial P_i(T_i, G_i)}{\partial G_i} &\leq 0, & \frac{\partial^2 P_i(T_i, G_i)}{\partial G_i^2} &\geq 0, \\ \frac{\partial^2 P_i(T_i, G_i)}{\partial G_i \partial G_j} &= 0, & \frac{\partial P_i(T_i, G_i)}{\partial T_i} &> 0, \\ \frac{\partial^2 P_i(T_i, G_i)}{\partial T_i^2} &\leq 0, & \frac{\partial^2 P_i(T_i, G_i)}{\partial T_i \partial T_j} &= 0; \end{aligned} \quad (1)$$

that is, $P_i(T_i, G_i)$ is convex and decreasing with respect to defensive investment G_i with diminishing marginal effectiveness, and is concave and increasing with respect to the attacker's effort T_i with diminishing marginal effectiveness, which means that the more defensive resources being invested to a target, the less likely it is that the target is attacked successfully. Conversely, the more resources that the attacker uses to attack a target, the higher the chance is that the target is attacked successfully. Note that in Equation (1), $i \neq j$. The probability of successful attack on target i ($P_i(T_i, G_i)$) is assumed to be impacted only by the attacker's resources to target i , T_i , and the defender's resources to target i , G_i ; that is, the cross partial derivatives ($\partial^2 P_i(T_i, G_i)/\partial G_i \partial G_j$, $\partial^2 P_i(T_i, G_i)/\partial T_i \partial T_j$) are assumed to be equal to zero.

We also assume

$$P_i(0, G_i) = \lim_{G_i \rightarrow \infty} P_i(T_i, G_i) = 0; \quad (2)$$

that is, the successful attack probability $P_i(T_i, G_i)$ of target i equals 0 either when no attack is launched or when the defender invests tremendous resources to protect the target i .

2.3. Defender's Optimization Model and Best Response

Subject to the budget constraint $\sum_{i=1}^n G_i \leq \mathcal{G}$, the goal of the defender is to find an optimal defensive resource allocation $G \equiv (G_1, G_2, \dots, G_n)$ minimizing the total expected loss $Z_G(T, G) = \sum_{i=1}^n D_i P_i(T_i, G_i)$. Note that,

Table 1. Notations Used in This Paper

Notation	Explanation
Model parameters	
n	Number of targets
A_i	The inherent defense level of target i , $\forall i = 1, 2, \dots, n$
$\mathcal{T}, \mathcal{G}$	The budgets of the attacker and the defender, respectively
B_i, D_i	The attacker's and the defender's valuation for the i th target, $\forall i = 1, 2, \dots, n$
β_i, α_i	Effectiveness coefficient of attack and defense investment to target i respectively, $\forall i = 1, 2, \dots, n$
λ_T^C, λ_G^C	Lagrangian multipliers for the attacker's and defender's budget constraints in simultaneous game (shadow prices)
λ_T^S, λ_G^S	Lagrangian multipliers for the attacker's and defender's budget constraints in sequential game (shadow prices)
$\mu_{T_i}^C, \mu_{G_i}^C$	Lagrangian multipliers for attacker's and defender's nonnegative constraints in simultaneous game (reduced cost), $\forall i = 1, 2, \dots, n$
$\mu_{T_i}^S, \mu_{G_i}^S$	Lagrangian multipliers for attacker's and defender's nonnegative constraints in sequential game (reduced cost), $\forall i = 1, 2, \dots, n$
Decision variables	
T_i	The attacker's investment in i th target, $\forall i = 1, 2, \dots, n$
T	$T \equiv (T_1, T_2, \dots, T_n)$
G_i	The defender's investment in i th target, $\forall i = 1, 2, \dots, n$
G	$G \equiv (G_1, G_2, \dots, G_n)$
Functions and equilibrium solutions	
$P_i(T_i, G_i)$	The probability of a successful attack to the target i
$Z_G(T, G)$	The objective of the defender
$Z_T(T, G)$	The objective of the attacker
$\hat{T}(G), \hat{G}(T)$	Attacker and defender's best response functions
(T^{C*}, G^{C*})	Nash equilibria for a simultaneous game
(T^{S*}, G^{S*})	Subgame perfect Nash equilibria for a sequential game

because of the assumption $\partial P_i(T_i, G_i)/\partial G_i \leq 0$, we have

$$\frac{\partial Z_G(T, G)}{\partial G_i} = \sum_{i=1}^n D_i \frac{\partial P_i(T_i, G_i)}{\partial G_i} \leq 0. \quad (3)$$

The defender's expected loss decreases in G . Therefore, to minimize the total expected loss $Z_G(T, G)$, the defender's budget constraint $\sum_{i=1}^n G_i \leq \mathcal{G}$ must be taut at $\sum_{i=1}^n G_i = \mathcal{G}$. Thus, the defender's optimization model is defined as follows:

$$\min_G Z_G(T, G) = \sum_{i=1}^n D_i P_i(T_i, G_i) \quad (4)$$

$$\text{subject to: } \sum_{i=1}^n G_i = \mathcal{G} \quad (5)$$

$$G_i \geq 0, \quad i = 1, 2, \dots, n, \quad (6)$$

where D_i represents the defender's i th target valuation.

Based on the defender's optimization problem in (4)–(6), the best response function of the defender is defined as

$$\hat{G}(T) \equiv \arg \min_{G \in W_G} Z_G(T, G), \quad (7)$$

where $W_G = \{\sum_{i=1}^n G_i = \mathcal{G}, G_i \geq 0\}$.

Proposition 1. First, the defender's optimization problem in (4)–(6) is convex. Second, there exist some constants (Lagrangian multipliers) $\mu_{G_1}, \dots, \mu_{G_n}$, and λ_G such that $\hat{G} = (\hat{G}_1, \dots, \hat{G}_n)$ solves the defender's optimization problem (4)–(6) globally if the following conditions in (8) are satisfied:

$$\begin{cases} -\sum_{k=1}^n D_k \frac{\partial P_k(T_k, \hat{G}_k)}{\partial G_i} - \lambda_G - \mu_{G_i} = 0, \\ \sum_{i=1}^n \hat{G}_i = \mathcal{G}, \\ \hat{G}_i \geq 0, \quad \forall i = 1, 2, \dots, n, \\ -\hat{G}_i \mu_{G_i} = 0, \quad \forall i = 1, 2, \dots, n, \\ \mu_{G_i} \geq 0, \quad \forall i = 1, 2, \dots, n. \end{cases} \quad \forall i = 1, 2, \dots, n, \quad (8)$$

Remark 1. Each of the Lagrangian multipliers λ_G and μ_{G_i} gives “shadow prices” associated with the budget constraint in (5) and each nonnegative constraint in (6); that is, if the defense budget $\mathcal{G}$ or the lower bound of the i th defense variable G_i is increased by a small

amount δ , the value of the optimal objective function will be increased by $\lambda_G \delta$ or reduced by $\mu_{G_i} \delta$, respectively. The “shadow prices” of the defender's non-negative constraint μ_{G_i} are also called reduced costs. The reduced cost equals 0 for $\hat{G}_i > 0$ in Equation (8) at an optimal point, at which the defender's marginal effects $\sum_{k=1}^n D_k (\partial P_k(T_k, \hat{G}_k)/\partial G_i)$ are the same for the targets with positive defense investments. Note that the optimization problem is convex, and the Karush–Kuhn–Tucker (KKT) conditions are both necessary and sufficient conditions. The solutions obtained in Proposition 1 are global optimal solutions.

2.4. Attacker's Optimization Model and Best Response

Similar to the defender's optimization problem, the attacker's objective is to find an optimal attack strategy $T \equiv (T_1, T_2, \dots, T_n)$ to maximize the total expected benefits, $\sum_{i=1}^n B_i P_i(T_i, G_i)$:

$$\max_T Z_T(T, G) = \sum_{i=1}^n B_i P_i(T_i, G_i) \quad (9)$$

$$\text{subject to: } \sum_{i=1}^n T_i = \mathcal{T}, \quad (10)$$

$$T_i \geq 0, \quad i = 1, 2, \dots, n. \quad (11)$$

Based on the attacker's maximization model (9)–(11), the attacker's best response function is defined as

$$\hat{T}(G) \equiv \arg \max_{T \in W_T} Z_T(T, G), \quad (12)$$

where $W_T = \{\sum_{i=1}^n T_i = \mathcal{T}, T_i \geq 0\}$.

Proposition 2. First, the attacker's optimization problem as defined in (9) is a convex problem. Second, there exist some constants (Lagrangian multipliers) $\mu_{T_1}, \mu_{T_2}, \dots, \mu_{T_n}$ and λ_T such that $\hat{T} = (\hat{T}_1, \hat{T}_2, \dots, \hat{T}_n)$ solves the attacker's optimization problem in (9)–(11) globally if the following conditions are satisfied:

$$\begin{cases} \sum_{k=1}^n B_k \frac{\partial P_k(\hat{T}_k, G_k)}{\partial T_i} - \lambda_T - \mu_{T_i} = 0, \\ \sum_{i=1}^n \hat{T}_i = \mathcal{T}, \\ \hat{T}_i \geq 0, \quad \forall i = 1, 2, \dots, n, \\ \mu_{T_i} \hat{T}_i = 0, \quad \forall i = 1, 2, \dots, n, \\ \mu_{T_i} \geq 0, \quad \forall i = 1, 2, \dots, n. \end{cases} \quad \forall i = 1, 2, \dots, n, \quad (13)$$

Remark 2. Note that in Section 3, the Lagrangian multipliers (λ_T, μ_{T_i}) of the attacker's budget and nonnegative constraints are denoted as λ_T^C and $\mu_{T_i}^C$, $\forall i = 1, 2, \dots, n$, for a simultaneous-move game, and λ_T^S and $\mu_{T_i}^S$, $\forall i = 1, 2, \dots, n$, for a sequential-move game. Since μ_{T_i} equals zero for any positive investments of the attacker $\hat{T}_i > 0$, similar to the cases of the defender, we note that the marginal effects of the attacker's investments $\sum_{k=1}^n B_k(\partial(P_k(\hat{T}_k, G_k))/\partial T_i)$ to each target are the same among all targets at the optimal point if $\hat{T}_i > 0$. Note that the optimization problem is convex, and the KKT conditions are both necessary and sufficient conditions. The solutions obtained in Proposition 2 are global optimal solutions.

3. Simultaneous-Move and Sequential-Move Games

3.1. Definition and Equilibrium Solution to the Simultaneous-Move Game

If both the defender and the attacker make their decisions without knowing the decision of the other at the time (e.g., because of secrecy), then a simultaneous-move game is appropriate to model the interactions. A Nash equilibrium solution to the simultaneous-move game, (T^C, G^C) , is defined as follows:

Definition 1. For a simultaneous-move game, we call a collection of strategies (T^C, G^C) a Nash equilibrium if and only if both Equations (14) and (15) are satisfied:

$$T^C = \hat{T}(G^C), \quad (14)$$

$$G^C = \hat{G}(T^C). \quad (15)$$

According to Definition 1, a Nash equilibrium solution to the simultaneous-move game, (T^C, G^C) , must optimize both the defender's and the attacker's models at the same time, which means that both the attacker's and defender's best response functions should be satisfied at the equilibrium point. Because the ranges of G and T are compact and convex, and Z_G and Z_T are continuous and convex and concave in G and T , respectively, a pure-strategy Nash equilibrium exists (Debreu 1952, Fan 1952, Glicksberg 1952). Thus, according to Propositions 1 and 2, if all the KKT conditions in Propositions 1 and 2 are satisfied, a Nash equilibrium solution (T^C, G^C) to the simultaneous-move game can be solved explicitly by applying Proposition 1 and Proposition 2 through simultaneously solving the KKT

conditions in (8) and (13). Note that the Lagrangian multipliers of the attacker's (defender's) budget and nonnegative constraints will be denoted as λ_T^C and $\mu_{T_i}^C$ (λ_G^C and $\mu_{G_i}^C$) ($\forall i = 1, 2, \dots, n$) in a simultaneous-move game.

3.2. Definition and Equilibrium Solution to a Sequential-Move Game

In some cases, the defender may purposefully or unavoidably disclose defense information by law, e.g., the airport security screening policy or the annual budget of the Department of Homeland Security (U.S. Department of Homeland Security 2017). Under these cases, a sequential-move game is more appropriate to model the interaction between the attacker and the defender. The goal of the defender is to minimize his total expected losses $Z_G(\hat{T}(G), G)$ by expecting the attacker's best response $\hat{T}(G)$ as defined in Equation (12); that is

$$\min_G Z_G = \sum_{i=1}^n D_i P_i(\hat{T}_i(G), G_i) \quad (16)$$

$$\text{subject to: } \sum_{i=1}^n G_i = \mathcal{G}, \quad (17)$$

$$G_i \geq 0, \quad i = 1, 2, \dots, n. \quad (18)$$

Definition 2. In a sequential-move game, we call a collection of strategies (T^{S^*}, G^{S^*}) a subgame-perfect Nash equilibrium (SPNE) if and only if both Equations (19) and (20) are satisfied:

$$G^{S^*} = \arg \min_{G \in W_G} Z_G(\hat{T}(G), G), \quad (19)$$

$$T^{S^*} = \hat{T}(G^{S^*}) = \arg \max_{T \in W_T} Z_T(T, G^{S^*}), \quad (20)$$

where the function $\hat{T}(\cdot)$ is the best response function of the attacker as defined in Equation (12).

Note that in this section, the Lagrangian multipliers of the attacker's (defender's) budget and nonnegative constraints will be denoted as λ_T^S and $\mu_{T_i}^S$ (λ_G^S and $\mu_{G_i}^S$) ($\forall i = 1, 2, \dots, n$) in a sequential-move game.

We also note that the problem in Equations (16)–(18) has the same convex feasible set as the problem in (4)–(6).

Proposition 3. For the defender's optimization problem in Equations (16)–(18), if there exist some constants (Lagrangian multipliers) $\mu_{G_1}^S, \mu_{G_2}^S, \dots, \mu_{G_n}^S$, and λ_G^S that

satisfy the following conditions, $G^{S^*} = (G_1^{S^*}, G_2^{S^*}, \dots, G_n^{S^*})$ is an optimal solution to the problem (16)–(18):

$$\begin{cases} -\sum_{k=1}^n D_k \left(\frac{\partial P_k}{\partial \hat{T}_k} \frac{\partial \hat{T}_k}{\partial G_i} + \frac{\partial P_k}{\partial G_i} \right) \Big|_{G=G^{S^*}} - \lambda_G^S - \mu_{G_i}^S = 0, \\ \quad \forall i = 1, 2, \dots, n, \\ \sum_{i=1}^n G_i^{S^*} = \mathcal{G}, \\ -G_i^{S^*} \leq 0, \quad i = 1, 2, \dots, n, \\ -G_i^{S^*} \mu_{G_i}^S = 0, \quad \forall i = 1, 2, \dots, n, \\ \mu_{G_i}^S \geq 0, \quad \forall i = 1, 2, \dots, n. \end{cases} \quad (21)$$

Remark 3. At SPNE, for any positive defender's investment $G_i^{S^*} > 0$ (to let $-G_i^{S^*} \mu_{G_i}^S = 0$ hold, we must have $\mu_{G_i}^S = 0$, $\forall i = 1, 2, \dots, n$), we note that the defender's total marginal effects $\sum_{k=1}^n D_k ((\partial P_k / \partial \hat{T}_k)(\partial \hat{T}_k / \partial G_i) + \partial P_k / \partial G_i) \Big|_{G=G^{S^*}}$ are the same among all targets and equal λ_G^S . Moreover, the defender's total marginal effects can be decomposed into two parts: the direct effect $\sum_{k=1}^n D_k (\partial P_k / \partial G_i) \Big|_{G=G^{S^*}}$, which shows how G_i directly influences P_k , and the indirect effect $\sum_{k=1}^n D_k ((\partial P_k / \partial \hat{T}_k)(\partial \hat{T}_k / \partial G_i)) \Big|_{G=G^{S^*}}$, in which G_i influences P_k by impacting $\hat{T}_k$, since the value of the indirect effect relies on the attacker's best response of investment, and the best response is a function of G_k . The indirect effect reveals how the marginal effect of the defender's total losses is influenced by the attacker's best response, and how the attacker's best response is impacted by the defender's investment.

4. Numerical Illustrations

4.1. Numerical Illustration for a Two-Target Protection Game

4.1.1. Successful Attack Probability Function. Although in Section 2.2 we used the probability of successful attack function to the i th target as a general form (e.g., ratio form, exponential form) contest success function, we will choose the ratio form CSFs for numerical illustration in this section. As long as all the assumptions and properties discussed in Section 2.2 are satisfied, the CSFs are always rational, and Proposition 3 is always valid no matter what form of CSF is adopted. In the illustration, we choose the ratio

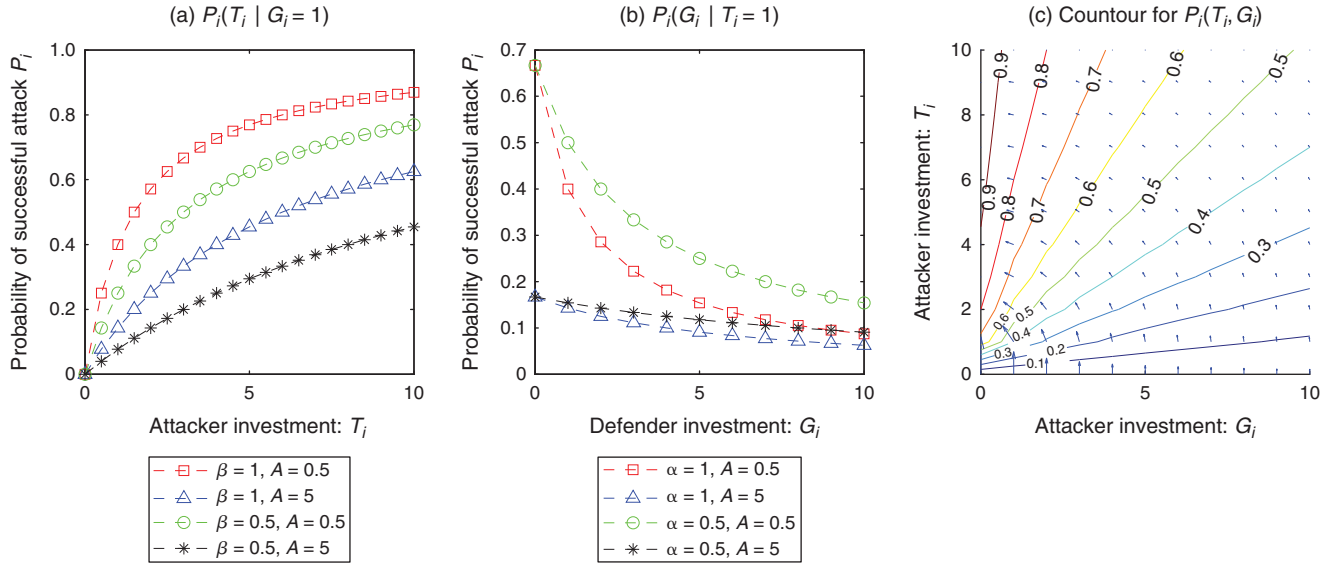
form CSFs (Hirshleifer 1989, Zhuang and Bier 2007, Hausken and Zhuang 2013):

$$P_i(T_i, G_i) = \frac{\beta T_i}{\beta T_i + \alpha G_i + A_i}. \quad (22)$$

In this ratio form CSFs, α and β are the effectiveness for the defender's and attacker's investments G_i and T_i , respectively. For example, to screen people or vehicles at borders that may be at potential risk of terrorist attacks, radiation portal monitors are frequently used by the defender. The portable portal monitors are much cheaper than large-scale portal monitors. However, portable portal monitors lack spectral identification capabilities. Meanwhile, large portal monitors may use multiple semiconductors (HPSe) or scintillators (NaI) as radiation detectors, which enable the gamma radionuclide identification capabilities that will allow for positive identification of radioactive isotopes for resolution of alarms (Cochran and McKinzie 2008). In this example, we can consider investment on large-scale portal monitors as high effective investment, which means that the same unit of such investment leads to a greater reduction on the successful attack probability when compared with a relatively low effective investment.

The inherent defense level A_i of target i is the level of the existing defense valued by monetary terms. The existence of such inherent defense indicates that even if the defender makes no investment on a certain target, the attacker may not always be able to attack the target successfully. For example, Coaffee (2016) shows that proper landscapes and climates in an area can inherently reduce the opportunity for successful attack. Estimating parameters (such as cost effectiveness of defense and inherent defense level) are generally difficult and not the focus of this paper; however, potential methods for parameter estimation include classical statistical analysis, a Bayesian approach and trend analysis, and the use of risk estimates rather than empirical data (e.g., simulation, expert opinion, fault trees, logic models, or system and reliability modeling; Jensen 2002, Biringer et al. 2007, Stewart and Mueller 2011). In estimating A_i for this particular case, we can use discounted carried-over defense from the previous period to estimate the inherent defense level.

Also, note that if we divide by A_i both the numerator and the denominator, the CSFs could be equivalently

Figure 1. (Color online) Illustration of a Probability of Successful Attack Function $P_i(T_i, G_i) = \beta T_i / (\beta T_i + \alpha G_i + A_i)$ 

rewritten as $P_i(T_i, G_i) = \beta(T_i/A_i) / (\beta(T_i/A_i) + \alpha(G_i/A_i) + 1)$, which only depends on the ratio T_i/A_i .

The probability of successful attack defined in Equation (22) satisfies all the conditions in Equations (1)–(2) and is illustrated in Figure 1. In particular, Figures 1(a)–1(c) shows that the successful attack probability increases in the attacker's effort T_i and decreases in the defender's effort G_i . Panels (a) and (b) show that the successful attack probability decreases as the i th target's inherent defense level A_i and the effectiveness of the defensive investment α increase, and the effectiveness of the attack investment β decreases.

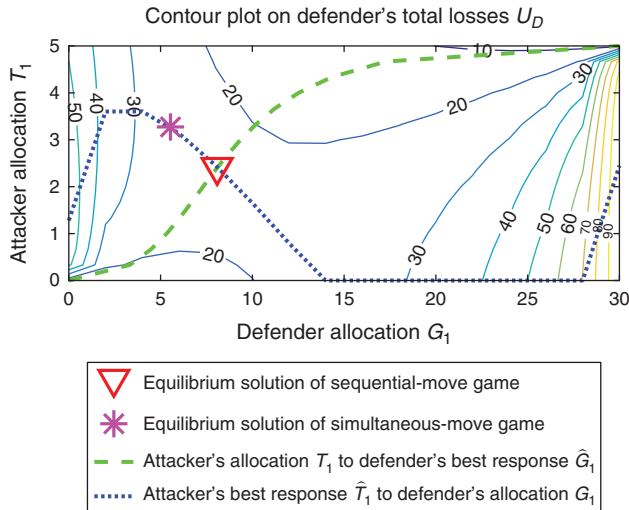
4.1.2. Numerical Illustrations for Best Response and Equilibrium Solution. By illustrating the problem using the two-target case and the ratio contest success function, $P_i(T_i, G_i) = \beta T_i / (\beta T_i + \alpha G_i + A_i)$, $i = 1, 2$, Equation (13) Proposition 2 becomes

$$\begin{cases} B_1 \frac{\beta(\alpha G_1 + A)}{(\beta \hat{T}_1 + \alpha G_1 + A)^2} + B_2 \frac{\beta(\alpha G_2 + A)}{(\beta(\mathcal{T} - \hat{T}_1) + \alpha G_2 + A)^2} - 2\lambda_T - \mu_1 - \mu_2 = 0, \\ \hat{T}_i \geq 0, \quad \forall i = 1, 2, \\ \mu_{T_i} \hat{T}_i = 0, \quad \forall i = 1, 2, \\ \mu_{T_i} \geq 0, \quad \forall i = 1, 2. \end{cases} \quad (23)$$

Now we set the parameters with baseline values of $\mathcal{T} = \$5$ million, $\mathcal{G} = \$30$ million, $B_1 = \$50$ million,

$B_2 = \$100$ million, $D_1 = \$150$ million, $D_2 = \$30$ million, $\beta_1 = \beta_2 = 1$, $\alpha_1 = \alpha_2 = 1$, and $A_i = 0.1$. First note that we must have $G_2 = \mathcal{G} - G_1$ and $T_2 = \mathcal{T} - T_1$. By varying the values of G_1 and T_1 , we get the contour plot of the defender's total losses $U_D(G_1, T_1)$. The dash line and dotted line in Figure 2 represent the attacker's best response $\hat{T}_1(G_1)$ and the defender's best response $\hat{G}_1(T_1)$, respectively. We see that the defender's best response $\hat{G}_1(T_1)$ increases as T_1 rises, because it is more efficient to defend a target with higher attacker's allocation. Moreover, we note that the attacker's best response $\hat{T}_1(G_1)$ shows a downward trend after the first rise and then rises again with the increase of defender's allocation G_1 . This is because when the defender chooses to allocate the majority of defensive resources on target 1, and allocates only a few resources on target 2, target 2 becomes vulnerable and easy to attack. In this scenario, the attacker does not need too many resources to successfully attack target 2. Therefore, the attacker will increase resource allocation to target 1. The intersection of those two best responses ("*") is the equilibrium solution of the simultaneous-move game. The equilibrium solution of the sequential-move game is also shown in Figure 2 ("▽"), which is the lowest point in the defender's utility contour plot that is intersected with the attacker's best response $\hat{T}_1(G_1)$. We note that the SPNE of the sequential-move game has a higher defender's allocation on target 1

Figure 2. (Color online) Contour Plot of the Defender’s Total Losses in a Two-Target Case and the Equilibrium Solutions of Both the Sequential-Move Game and Simultaneous-Move Game



and a lower defender’s total losses (\$20.88 million) than the Nash equilibrium of the simultaneous-move game (\$22.60 million), which indicates that there is a first-mover advantage when the defender moves first by disclosing defensive information.

4.2. Numerical Illustration for a Multiple-Infrastructure Protection Game

4.2.1. Parameter Setting. The numerical example for a multiple-infrastructure game, where the attacker and defender have different infrastructure valuations, is illustrated in this section, with baseline values of model parameters $\mathcal{G} = \$30$ million, $\mathcal{T} = \$5$ million, $n = 10$, $\beta = 2$, $\alpha = 2$, and $A_i = 0.1$. The attacker’s and the defender’s valuations of the i th infrastructure, B_i , and D_i , are generally different and listed in Table 2 for 10 critical infrastructures within a state: public building, water supply, military bases, rail system, ports, health system, national leaders, air travel, Internet, and telecommunication.

4.2.2. Numerical Illustrations for Equilibrium Solution of the Simultaneous-Move Game. We solve the equilibrium solution for both the simultaneous and sequential move games by using an algorithm provided by LINGO software.

Based on the baseline values of the parameter provided in Section 4.2.1, the attacker’s and the defender’s

Table 2. Attacker’s and Defender’s Valuations for 10 Critical Infrastructures

#	Infrastructure	Attacker’s valuation B_i (\$ million)	Defender’s valuation D_i (\$ million)
1	Public building	10	30
2	Water supply	10	150
3	Military bases	1	40
4	Rail system	10	15
5	Ports	15	100
6	Health system	20	70
7	National leaders	100	1,000
8	Air travel	50	50
9	Internet	2	75
10	Telecommunication	20	150

equilibrium allocation strategies, expected payoffs, marginal effects, and probability of successful attack are provided in Table 3.

Table 3 shows that all targets are both attacked and defended in the simultaneous-move game. The attacker attacks target 8, air travel, with most of his resources (\$2.176 million), while the defender allocates most of the defense resource to her most valuable target, target 7, national leaders. The attacker’s expected benefit and the defender’s total expected loss equal to $\sum_{i=1}^{10} P_i(T_i, G_i)B_i = \38.50 million and $\sum_{i=1}^{10} P_i(T_i, G_i)D_i = \136.27 million, respectively. From the marginal effects of the attacker’s budget λ_T^C and the defender’s budget λ_G^C , we note that increasing attack resources by \$1 million would increase attacker benefits by \$5.54 million, and increasing the defensive budget by \$1 million would reduce defender losses by \$3.79 million. Though among all targets, the expected payoffs of the attacker (P_iB_i) and defender (P_iD_i) are different, as predicted by Proposition 2, Table 3 shows that at the equilibrium, the attacker’s marginal effects of each attacked target ($B_i(\partial P_i(T_i, G_i)/\partial T_i)$) are the same across the targets and equal to λ_T^C (\$−3.79 million), and the defender’s marginal effects of each defended target ($D_i(\partial P_i(T_i, G_i)/\partial G_i)$) are the same as well and equal to λ_G^C (\$5.54 million).

According to the results, it is interesting that the attacker would spend the most on target 8 even though it is not the most valuable target to them. We notice that target 8 has a second-highest valuation to the attacker and a relatively low valuation to the defender (compared to \$1,000 million for target 7). Since target 7 has an extremely high valuation, the defender will tend to

Table 3. Equilibrium Solutions, Successful Attack Probability, Expected Payoffs, and Marginal Effects in the Simultaneous Game (\$ Million)

Target #	Equilibrium solutions		Probability	Expected payoffs		Marginal effects	
	$T_i^{C^*}$	$G_i^{C^*}$	$P_i(T_i^{C^*}, G_i^{C^*})$ (%)	$P_i B_i$	$P_i D_i$	$B_i(\partial P_i(T_i, G_i)/\partial T_i)$	$D_i(\partial P_i(T_i, G_i)/\partial G_i)$
1	0.273	1.097	18.57	1.86	5.57	5.54	−3.79
2	0.075	1.551	4.35	0.43	6.52	5.54	−3.79
3	0.003	0.075	1.69	0.02	0.67	5.54	−3.79
4	0.388	0.752	31.29	3.13	4.69	5.54	−3.79
5	0.228	2.127	9.29	1.39	9.29	5.54	−3.79
6	0.493	2.426	16.33	3.27	11.43	5.54	−3.79
7	1.081	15.71	6.40	6.40	64.00	5.54	−3.79
8	2.176	3.083	40.60	20.30	20.30	5.54	−3.79
9	0.006	0.248	1.69	0.03	1.27	5.54	−3.79
10	0.276	2.931	8.35	1.67	12.52	5.54	−3.79
Total	5.00	30.00		38.50	136.27		
Marginal effect for attacker budget λ_T^C						5.54	
Marginal effect for defender budget λ_G^C						−3.79	

allocate the majority of its budget to target 7 rather than to other targets. Therefore, it is rational that the attacker would spend most on target 8, because of its high valuation and relatively low defense level.

4.2.3. Numerical Illustration for SPNE Solution of Sequential-Move Game. Based on the baseline values of parameters provided in Section 4.2.1 and applying Proposition 3, the defender's and the attacker's SPNE equilibrium strategies in resource allocation, expected payoffs, marginal effects, and probability of successful attack are provided in Table 4.

From Table 4, we find that the defender defends his most valuable target, number 7, national leaders, with most of his resources, $G_7^{S^*} = \$18.746$ million. Comparing with the results from the simultaneous-move game in Table 3, the attacker chooses only a limited set of targets (targets 1, 4, 5, 6, 8, and 10) to attack, instead of attacking all of the targets. However, the defender defends all targets. We note that the attacker's expected benefits are $\sum_{i=1}^{10} P_i(T_i, G_i) \cdot B_i = \46.99 million and the defender's total expected losses are $\sum_{i=1}^{10} P_i(T_i, G_i) \cdot D_i = \108.09 million, where both the attacker and defender have better expected payoffs in the sequential game model than in the simultaneous game.

The attacker's benefit $P_i B_i$ and defender's loss $P_i D_i$ are not the same across target i , $\forall i = 1, 2, \dots, n$. For marginal effects of the attacker's and defender's investments to each target, we note that increasing attack resources by \$1 million would increase the attacker's

benefits by \$5.31 million (less than the \$5.54 million in the simultaneous game), and increasing defensive resources by \$1 million would reduce the defender's losses by \$4.20 million (more than \$3.39 million in the simultaneous game). Targets 2, 3, 7, and 9 would not be attacked ($T_i^{S^*} = 0$), making the attacker's marginal effects zero. For the defender, the total marginal effects are positive and equal (\$−4.20 million) for all targets.

By comparing Tables 3 and 4, we note that there is a first mover advantage for the defender in terms of lower expected losses in the sequential game (\$108.09 million, compared to \$136.27 million in the simultaneous game), which suggests that defensive allocation should be disclosed rather than kept secret.

Note that to generalize the model, we assume that the attacker has the option of launching multiple attacks to multiple targets. However, the result shows that the attacker may choose to not attack some of the targets because of limited budget (see Table 4). It is also possible that only a single target would be attacked when the attacker's budget is really tight.

4.2.4. Sensitivity Analysis. To better understand how the attacker's and the defender's equilibrium strategies and expected payoffs depend on the model parameters (such as cost effectiveness of the attack and defensive resource allocation, inherent defense levels of targets, and attacker's and defender's budgets), we conducted sensitivity analyses using the same set of baseline values as in Section 4.2.1 for both the sequential-move and simultaneous-move games.

Table 4. Equilibrium Solutions, Successful Attack Probability, Expected Payoffs, and Marginal Effects in the Sequential Game (\$ Million)

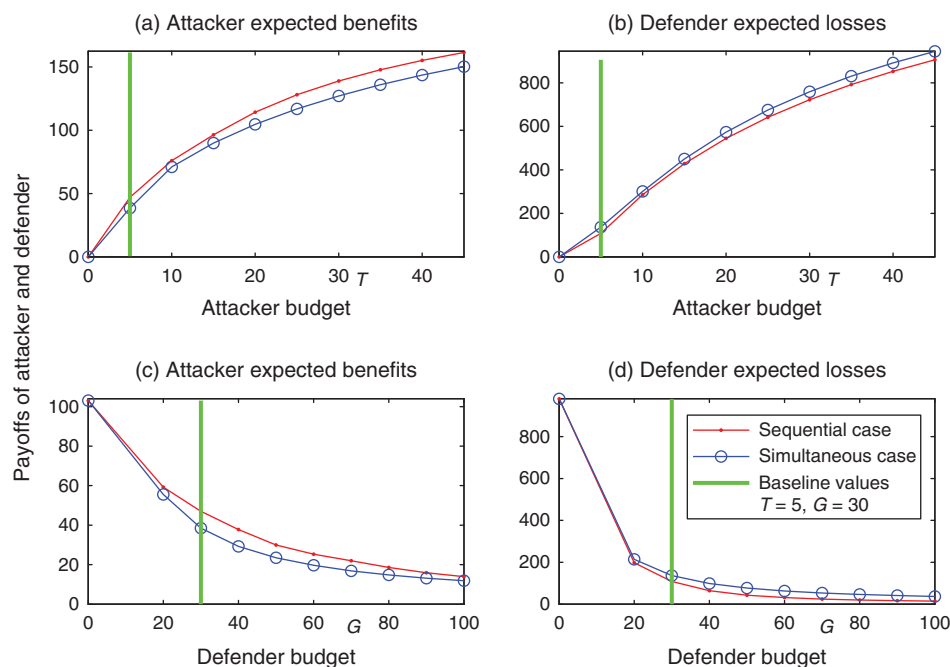
Target #	Equm. sol.		Probability	Exp. payoffs		Marginal effects	
	T_i^{S*}	G_i^{S*}	$P_i(T_i^{S*}, G_i^{S*})$ (%)	$P_i B_i$	$P_i D_i$	$B_i(\partial P_i(T_i, G_i)/\partial T_i)$	$\sum_{k=1}^n D_k((\partial P_k/\partial T_k)(\partial T_k/\partial G_i) + \partial P_k/\partial G_i)$
1	0.450	0.591	39.44	3.94	11.83	5.31	−4.20
2	0.000	1.785	0.00	0.00	0.00	0.00	−4.20
3	0.000	0.088	0.00	0.00	0.00	0.00	−4.20
4	0.471	0.406	48.21	4.82	7.23	5.31	−4.20
5	0.419	1.798	18.08	2.71	18.08	5.31	−4.20
6	0.874	1.418	36.54	7.31	25.58	5.31	−4.20
7	0.000	18.746	0.00	0.00	0.00	0.00	−4.20
8	2.355	2.152	51.12	25.56	25.56	5.31	−4.20
9	0.000	0.277	0.00	0.00	0.00	0.00	−4.20
10	0.432	2.739	13.21	2.64	19.81	5.31	−4.20
Total	5.00	30.00		46.99	108.09		
Marginal effect for attacker budget λ_T^S						5.31	
Marginal effect for defender budget λ_G^S						−4.20	

Figure 3, (a) and (b), shows the changes in the attacker's expected benefits and the defender's expected losses while keeping the defender's budget constant and varying the attacker's budget. Both the attacker's expected benefits and the defender's expected losses increase with the attacker's budget. Similarly, Figure 3, (c) and (d), shows that both the

defender's expected losses and the attacker's expected benefits decrease when the defender's budget increases while keeping the attacker's budget constant and varying the defender's budget.

Figure 3 shows that the defender has fewer expected losses in the sequential game model than in the simultaneous game. This is known as the first-mover

Figure 3. (Color online) Comparisons of the Defender's and Attacker's Expected Payoffs in Simultaneous and Sequential Games When the Attacker's and Defender's Budgets Vary



advantage, which is significantly large when the attacker's budget is large or when the defender's budget is moderate.

In the sequential-move game, when the attacker's total budget is increased by \$1 million in the baseline case ($\mathcal{T} = \$5$ million, $\mathcal{G} = \$30$ million), the defender's total loss will increase from \$108.08 million to \$146.97 million. However, in the simultaneous-move game, the defender's total loss will increase from \$102.69 million to \$123.27 million when the attacker's total budget is increased by \$1 million.

In practice, although the extraneous variables could not be controlled for as in lab experiments, there is evidence that shows that a small increment on the cost of a terrorist attack could lead to a lot more losses for the defender. For example, the 1993 World Trade Center bombing cost Al Qaeda \$18,000 (Llussá and Tavares 2011) and resulted in \$300 million in property damage (Caram 2001), while the September 11, 2001, attacks cost Al Qaeda \$500,000 (Little et al. 2007) and caused property damage over \$10 billion (Feeney et al. 2005). The increased cost of \$482,000 for terrorists led to an increment on the defender's losses of more than \$9.5 billion.

4.2.5. Regression Analysis. In this section, we use linear regression to study the correlations between the defender's and the attacker's equilibrium strategies (G_i^C and T_i^C for the simultaneous game and G_i^S and T_i^S for the sequential game, as presented in Sections 4.2.2 and 4.2.3) and their target valuations, B_i and D_i , for each target i with the inputs from Tables 3 and 4 as the parameter setting for the specific case shown in Section 4.2.1. The results are listed below:

$$T_i^C = 0.0440763 B_i - 0.0033422 D_i + 0.0123699, \quad (24)$$

(1.91e-06)**** (1.38e-05)**** (0.831)**

$$G_i^C = 0.009547 B_i + 0.062604 D_i - 0.812018, \quad (25)$$

(0.000472)**** (3.7e-05)**** (0.626754)*

$$T_i^S = 0.0475755 B_i - 0.0049343 D_i + 0.1967579, \quad (26)$$

(2.70e-08)**** (2.37e-08)**** (0.0005)****

$$G_i^S = 0.0435862 B_i + 0.0148941 D_i - 0.5395517. \quad (27)$$

(0.00026)**** (8.09e-08)**** (0.00241)****

The multiple R^2 values of the regressions in Equations (24)–(27) are equal to 0.961, 0.991, 0.988, and 0.997 respectively, which indicates that the regression results are highly significant. The numbers in parentheses under the coefficients are the corresponding p values of the standard error of coefficient estimates

(Sellke et al. 2001). The asterisks in Equations (24)–(27) indicate significance levels as follows: * $p \geq 0.1$, ** $0.1 > p \geq 0.05$, *** $0.05 > p \geq 0.01$, and **** $p < 0.01$. The p values under B_i and D_i are all less than 0.01, which implies that there is significant correlation between B_i and D_i at the equilibrium solutions for both the simultaneous and sequential games.

From Equations (24)–(27), we observe that an equilibrium strategy of an attacker is positively correlated with his target valuations and negatively correlated with the defender's, while the optimal strategy of the defender is positively correlated with both the attacker's and the defender's target valuations.

4.3. Model Illustration in Multiple-Urban-Area Protection Game

4.3.1. Parameters Setting. In this section, we illustrate the model by using the data of expected property losses and the FY 2004 Urban Area Security Initiative (UASI) budget for the top 10 urban areas in the United States (Willis et al. 2005) shown in Table 5. We assume that the target valuations of the defender and the attacker are the same ($v = V$) in this section. Thus, the defender's expected losses are equal to the attacker's expected benefits. The defender's budget equals the budget requirement received by the DHS in 2012 (U.S. Department of Homeland Security 2017), which is $\mathcal{G} = \$268$ million. Based on data from the TFTP program (Roth et al. 2004), we set the attacker's budget as $\mathcal{T} = \$30$ million. The baseline values of other model parameters are $A = 0.1$ and $\alpha = \beta = 1$.

Table 5. Expected Property Losses, Expected Fatalities, and FY 2004 UASI Budget Allocations for 10 Highest Losses Urban Areas (\$ Million)

#	Urban areas	Target valuation D_i	FY 2004 UASI allocations
1	New York City	413.0	47
2	Chicago	115.0	34
3	San Francisco	57.0	26
4	Washington, D.C.	36.0	29
5	Los Angeles–Long Beach	34.0	40
6	Philadelphia, PA–NJ	21.0	23
7	Boston, MA–NH	18.0	19
8	Houston	11.0	19
9	Newark	7.3	15
10	Seattle–Bellevue	6.7	16
	Total	719	268

Table 6. Equilibrium Solutions, Successful Attack Probability, Expected Payoffs, and Marginal Effects in the Simultaneous Game (\$ Million)

Target #	Equilibrium solutions		Probability	Expected payoffs	Marginal effects	
	$T_i^{C^*}$	$G_i^{C^*}$	$P_i(T_i^{C^*}, G_i^{C^*})$ (%)	$P_i B_i = P_i D_i$	$B_i(\partial P_i(T_i, G_i)/\partial T_i)$	$D_i(\partial P_i(T_i, G_i)/\partial G_i)$
1	1.69	187.43	0.9	3.69	2.16	−0.70
2	0.14	38.69	0.3	0.40	2.16	−0.70
3	5.64	8.36	40.0	22.80	2.16	−0.70
4	3.05	3.88	43.3	15.60	2.16	−0.70
5	3.86	14.23	21.2	7.21	2.16	−0.70
6	0.98	3.44	21.8	4.57	2.16	−0.70
7	1.45	4.33	24.7	4.44	2.16	−0.70
8	8.04	4.86	61.8	6.80	2.16	−0.70
9	2.79	1.38	65.3	4.77	2.16	−0.70
10	2.37	1.40	61.2	4.10	2.16	−0.70
Total	30.00	268.00		74.39		
Marginal effect for attacker budget λ_T^C					2.16	
Marginal effect for defender budget λ_G^C					−0.70	

4.3.2. Equilibrium Solutions. The equilibrium solutions to the simultaneous game and sequential game are showed in Tables 6 and 7, respectively.

The equilibrium of the simultaneous game in Table 6 shows all targets are defended and attacked. In particular, the defender assigns the most resources to her most valuable target, New York City. The attacker attacks the middle and lower middle valuation level targets with most of his attack efforts (target 3, San Francisco; target 4, Washington, D.C.; target 5, Los Angeles; and target 8, Houston), instead of the cities

with the highest evaluation. Target 8 (Houston), target 9 (Newark), and target 10 (Seattle) are the top three risky targets with more than 60% probability to be attacked successfully. The defender's expected losses are \$74.39 million. From the marginal effects of attacker and defender resource allocation, we note that increasing defensive resources by \$1 million would decrease the defender's expected losses by \$0.70 million, and increasing attacker resources by \$1 million would increase the attacker's expected benefits by \$1.24 million. We also observe that the "small probabilities"

Table 7. Equilibrium Solutions, Successful Attack Probability, Expected Payoffs, and Marginal Effects in the Sequential Game (\$ Million)

Target #	Equilibrium solutions		Probability	Expected payoffs	Marginal effects	
	$T_i^{S^*}$	$G_i^{S^*}$	$P_i(T_i^{S^*}, G_i^{S^*})$ (%)	$P_i B_i = P_i D_i$	$B_i(\partial P_i(T_i, G_i)/\partial T_i)$	$\sum_{k=1}^n D_k((\partial P_k/\partial T_k)(\partial T_k/\partial G_i) + \partial P_k/\partial G_i)$
1	19.54	165.44	10.6	43.60	1.24	−0.24
2	3.28	35.74	8.4	9.19	1.24	−0.24
3	1.67	18.34	8.3	4.73	1.24	−0.24
4	1.18	12.14	8.8	3.17	1.24	−0.24
5	1.03	11.00	8.5	2.87	1.24	−0.24
6	0.66	6.89	8.6	1.81	1.24	−0.24
7	0.93	5.87	13.5	2.43	1.24	−0.24
8	0.63	5.51	10.0	1.10	1.24	−0.24
9	0.66	3.59	15.2	1.11	1.24	−0.24
10	0.42	3.49	10.5	0.70	1.24	−0.24
Total	30.00	268.00		71.17		
Marginal effect for attacker budget λ_T^S					1.24	
Marginal effect for defender budget λ_G^S					−0.24	

(e.g., 0.9% and 0.3% for targets 1 and 2, respectively) occur when the defender invests far more than the attacker on a certain target, which means that with the great effort made by the defender, the potential that targets will be successfully attacked could be significantly reduced.

From Table 7, we note that all targets are also defended and attacked in the sequential game model. Both the attacker and defender allocate most of their resources to their most valuable targets, which is target 1, New York City, in this example. Newark has the highest probability to be attacked successfully, and New York City would suffer the highest expected loss from an attack. The marginal effect of the defender's budget ($\lambda_G^S = -0.24$) implies that increasing defensive resources by \$1 million could reduce the defender's losses by \$0.24 million.

Comparing the results in Tables 6 and 7, we note that the expected losses of the defender are \$74.39 million in the simultaneous game, which is higher than the losses in the sequential game, \$71.17 million. This is consistent with the conclusion we drew from Section 4.2.3.

4.3.3. Regression Analysis. Based on the equilibrium solutions in Table 6 (G_i^C, T_i^C) and Table 7 (G_i^S, T_i^S), we conducted a regression analysis to study the relationships between the equilibrium solutions and the target valuations for the simultaneous-move and sequential-move games using the parameter settings and the equilibrium solutions of the specific case presented in Sections 4.3.1 and 4.3.2. Since we assume that the defender's target valuation D_i and the attacker's target valuation B_i are the same in this section, we denote the target valuations as D_i in our regression analysis. The results of the linear regression are shown as follows:

$$T_i^C = -0.005114 D_i + 3.368710 \quad (R^2 = 0.07269, p\text{-value} = 0.4512), \quad (28)$$

(0.45124)* (0.00539)****

$$G_i^C = 0.009547 D_i - 6.31111 \quad (R^2 = 0.988, p\text{-value} = 5.704e-09), \quad (29)$$

(5.7e-09)**** (0.0343)***

$$T_i^S = 0.046894 D_i - 0.371667 \quad (R^2 = 0.9841, p\text{-value} = 1.762e-08), \quad (30)$$

(1.76e-08)**** (0.237)*

$$G_i^S = 0.399068 D_i - 1.891981 \quad (R^2 = 0.9953, p\text{-value} = 1.368e-10). \quad (31)$$

(1.37e-10)**** (0.196)***

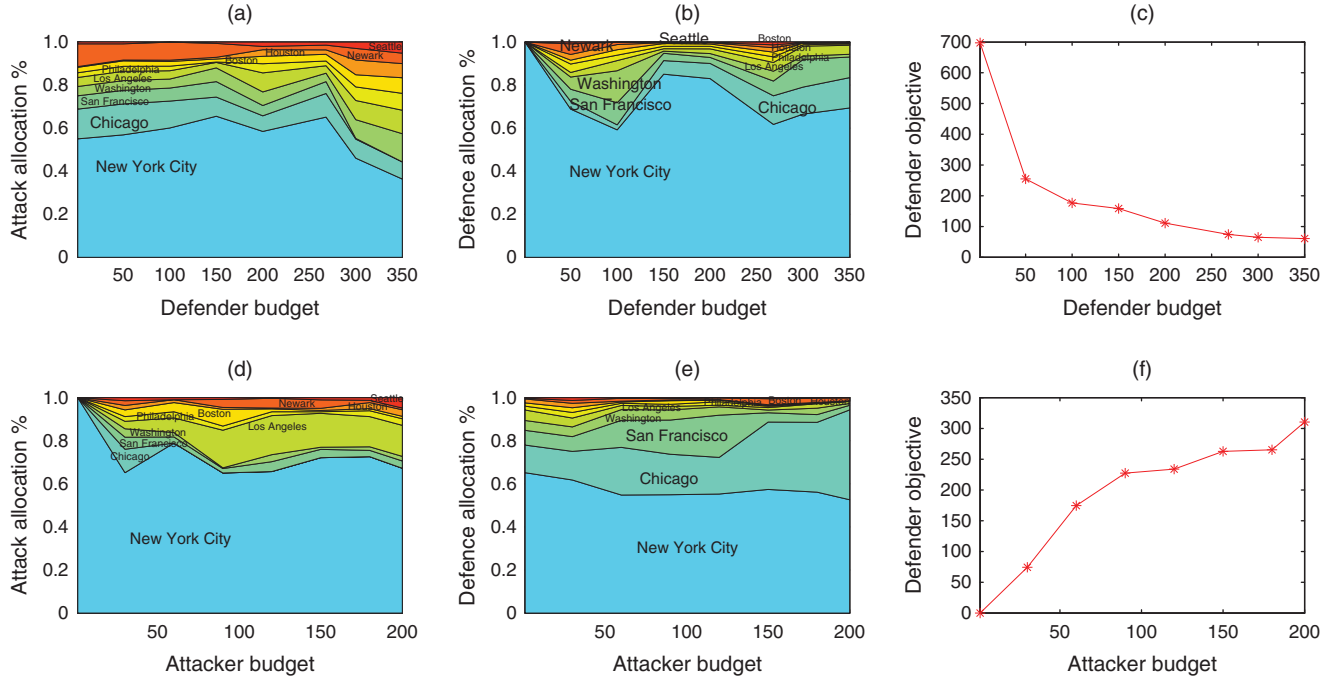
Similar to Equations (24)–(27), the asterisks in Equations (28)–(31) indicate significance levels as follows: * $p \geq 0.1$, ** $0.1 > p \geq 0.05$, *** $0.05 > p \geq 0.01$, and **** $p < 0.01$. According to the statistical analysis results (R^2 and p -value), we find that only the attacker's equilibrium strategy is not significantly correlated to the target valuation in the simultaneous case of a multiple-urban-area protection game. Note that this is not surprising, since we assume that the attacker's and defender's target valuations are the same for all targets, which may cancel some correlation effects. We observe that both the attacker's and defender's equilibrium strategies are highly positively correlated with the target valuation in the sequential-move game as well as the defender's equilibrium strategy in the simultaneous-move game. These results are consistent with the results in Section 4.3.3.

4.3.4. Sensitivity Analysis. Figure 4 shows the sensitivity analysis of the attacker's and defender's resource allocations and the defender's expected losses while the attacker's and defender's budget constraints change in a sequential game. Figure 4, (a) and (d), shows that the attacker would switch his attack effort from the most valuable target (target 1, New York City) to other targets. The attacker switches a few attack efforts from the most valuable target to the less valuable targets as the defender's budget increases. Figure 4, (b) and (e), shows that the defender will enlarge the percentage of the resources to his second most valuable target (target 2, Chicago) when either the defender's or the attacker's budget increases. Panels (c) and (f) show that the defender's expected losses decrease as the defender's budget increases with a diminishing marginal return, and increase as the attacker's budget increases with diminishing marginal returns.

We study how the defender's and the attacker's optimal strategies change under different inherent defense levels in both simultaneous and sequential games, which are shown in Figures 5 and 6. Note that the heights of the bars in the figures represent the percentages of resource allocation to the targets.

Figure 5, (a) and (c), shows that when the targets' inherent defense levels are low ($A = 0.01$), at equilibrium solutions, the defender would allocate most of her resources to her highly valuable targets (target 1, New York City, and target 2, Chicago), while the attacker would attack his most valuable target (target 1,

Figure 4. (Color online) Comparisons of Defender’s and Attacker’s Resource Allocation and Defender’s Expected Loss as Functions of Defender’s and Attacker’s Budget Constraints in Sequential Game



New York City) with most of his resources. Figure 5 also shows that both the attacker and the defender evenly allocate their resources to all the targets when inherent defense levels are sufficiently high.

Figure 6 shows how the attacker’s and the defender’s equilibrium resource allocations change in equilibrium with two inherent defense levels in a sequential game. Figure 6 shows that the resource allocations of the attacker and the defender to each target are similar in low and high target inherent defense level cases. Both players allocate most of their resources to their most valuable targets when the inherent defense levels are low, while they allocate their resources evenly to all the targets when the inherent defense levels are sufficiently high.

4.3.5. Multiple-Target Model vs. Single-Target Model.

For the attacker’s optimization model in Section 2.4, the attacker is assumed to be able to attack multiple targets. To examine the utility of the new model, we study what the consequence is if the defender believes that only one target will be attacked, but the attacker launches multiple attacks; that is, the attacker’s true

MT models and attacker’s models in defender’s belief (ST) are as follows:

$$(\text{Belief: ST}) \quad \underset{T'}{\text{maximize}} \quad Z_T(T', G) = \sum_{i=1}^n B_i P_i(T'_i, G_i)$$

$$\text{subject to: } T'_i = Z_i \times \mathcal{T},$$

$$\sum_{i=1}^N Z_i = 1, \quad Z_i = 0 \text{ or } 1,$$

$$T'_i \geq 0, \quad \forall i = 1, 2, \dots, n,$$

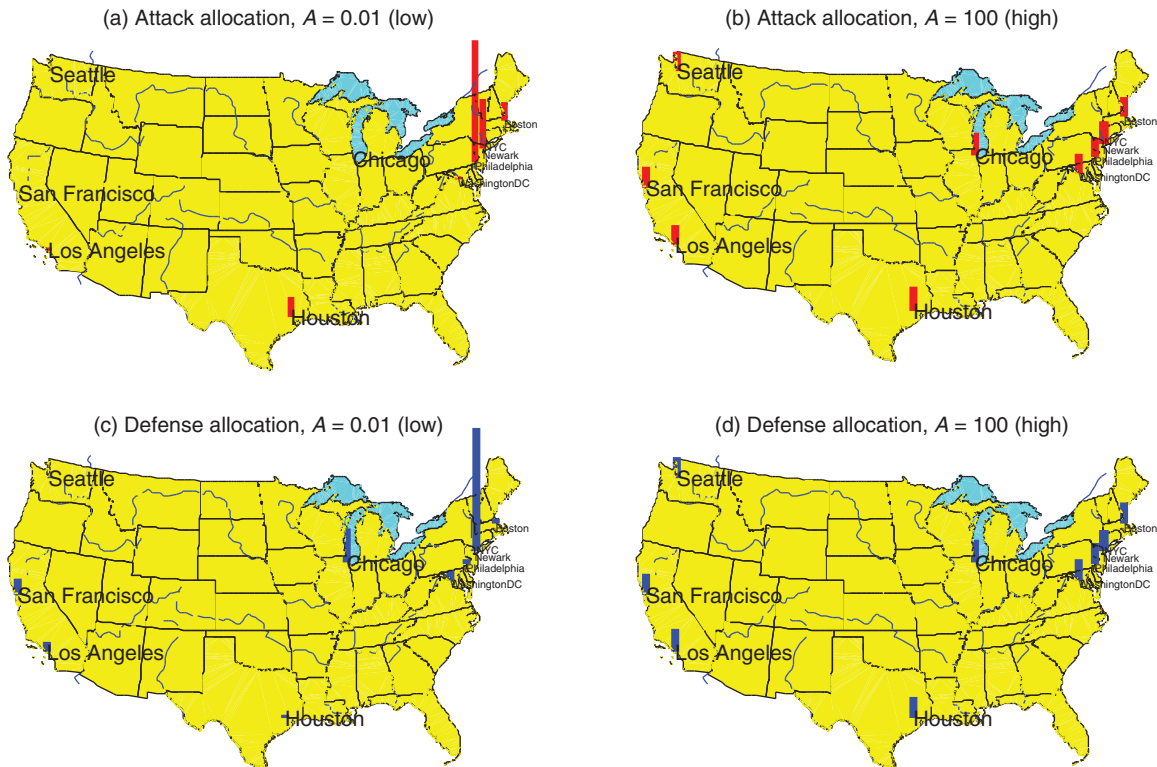
$$(\text{Truth: MT}) \quad \underset{T}{\text{maximize}} \quad Z_T(T, G) = \sum_{i=1}^n B_i P_i(T_i, G_i)$$

$$\text{subject to: } \sum_{i=1}^N T_i = \mathcal{T}$$

$$T_i \geq 0, \quad \forall i = 1, 2, \dots, n,$$

where Z_i is the binary variable and T'_i is the artificial variable assumed by the defender in the ST attack model. After solving the defender’s strategy in the ST model, the true attacker’s MT best response is used to calculate the players’ payoffs in the ST model.

We compare the attacker’s and defender’s losses/benefits in both the MT and ST models in both the sequential-move game and the simultaneous-move

Figure 5. (Color online) Comparisons of Attacker's and Defender's Resource Allocations on Different Inherent Defense Levels in the Simultaneous Game

game for both the multiple-infrastructure data set (data 1) and the multiple-urban-area data set (data 2). The results are shown in Table 8.

Using data 1, Table 8 shows that if the defender assumes that the attacker will only attack a single target but in reality the attacker attacks multiple targets, the risk would be underestimated. In particular, Table 8 also shows that the defender would suffer 4.5 (485.70/108.08) times higher losses in the sequential-move game and 2.5 (336.00/136.36) times higher losses in the simultaneous-move game by assuming a single target attack instead of the multiple attack cases. Similarly, using data 2, the defender would suffer 2.4 (172.52/71.17) times higher losses in the sequential-move game and 2.2 (163.90/74.39) times higher losses in the simultaneous-move game by assuming a single target attack instead of multiple attacks.

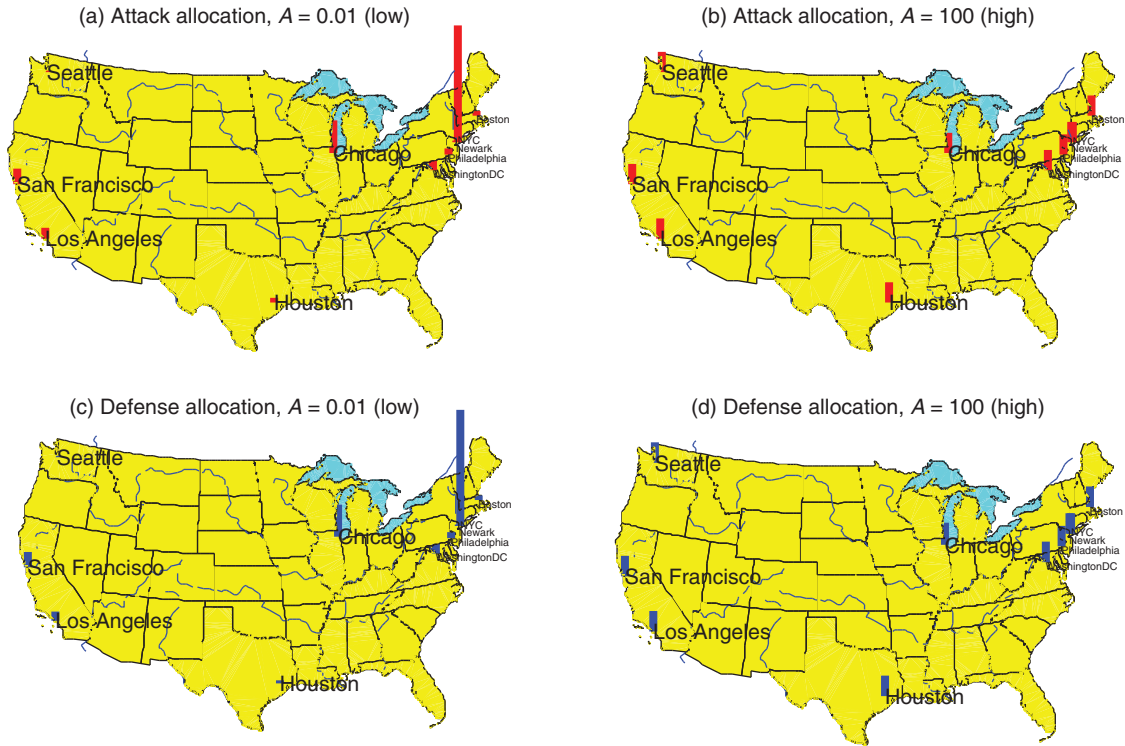
The above results justify the importance of the new model proposed in this paper. Because of the defender's false belief, the attacker enjoys a slight increase in expected benefits, which are 1.12 (52.77/46.98)

times higher in the sequential-move game and 1.13 (43.56/38.51) times higher in the simultaneous-move game using data 1, and the defender suffers 2.4 (172.52/71.17) times higher losses in the sequential-move game and 2.2 (163.90/74.39) times higher losses in simultaneous-move game using data 2. Note that this is the result from the illustration example under the baseline value setting for the model parameters.

5. Conclusions and Future Research Directions

To our knowledge, although in practice both the attacker and the defender have budget constraints, no previous research has explicitly considered both attacker's and defender's recourse allocation problems under budget constraints in a game-theoretical setting. This paper studies the attack effort and defensive effort in a continuous manner rather than a discrete one (attack or not attack, defend or not defend). The equilibria of both sequential and simultaneous games are obtained by solving constrained nonlinear

Figure 6. (Color online) Comparisons of Attacker’s and Defender’s Resource Allocations on Different Inherent Defense Levels in the Sequential Game



optimization problems. Two data sets are provided to illustrate the models.

We conduct extensive sensitivity analyses on the effects of the defender’s and attacker’s budgets, target valuations, and cost effectiveness of investments, and the inherent defense levels of the various targets. We find that the attacker’s optimal strategy is positively correlated with his target valuations but negatively correlated with the defender’s, while the defender’s optimal strategy is positively correlated with both the attacker’s and the defender’s target valuations.

Table 8. Compare the Payoffs of the Attacker and the Defender Between the Multiple Targets Attack Model and the One Target Attack Model (Unit: \$ Million)

Game type	SG		CG	
	MT	ST	MT	ST
Defender expected losses (data 1)	108.08	485.70	136.36	336.00
Attacker expected payoffs (data 1)	46.98	52.77	38.51	43.56
Attacker/defender expected payoffs (data 2)	71.17	172.52	74.39	163.90

Note. SG, sequential-move game; CG, simultaneous-move game.

Interestingly, we find that the attacker is less concentrated on attacking the most valuable target and spreads the resources to attack more targets as his budget increases. We find that the defender has less expected damage in the sequential game model than in the simultaneous one. From the analysis of the inherent defense levels of targets, we note that both players allocate most of their resources to their most valuable targets if all targets’ inherent defense levels are low, while they evenly distribute their resources to all targets if the targets’ inherent defense levels are high in both sequential and simultaneous games.

To show the utility of the new model, we compare the results of two models: the model with multiple targets attacked and the model in which the defender assumes that only a single target will be attacked when actually there could be multiple targets attacked. Our results show that the defender will suffer a 2.2–4.5 times higher expected loss if they assume that the attacker will only attack one target.

For simplicity, this paper assumes one defender and one attacker, as well as complete information and

common knowledge of the target valuations, budgets, and the contest success function. Also, we do not consider the interdependencies between different targets, uncertainty of the defender or attacker, and bounded rationality of players for simplicity. We could extend the model by considering multiple defenders (e.g., different governments/countries), multiple attackers (e.g., different terrorism organizations), incomplete budget and target valuation information, interdependencies between targets, defender or attacker uncertainty, or bounded rationality of players, as well as the different beliefs of the forms of the contest success function.

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Appendix A

A.1. Proof of Proposition 1

First, to prove the defender's optimization model in (4) is a convex problem, we need to show her objective function is concave and the constraint set is convex (Bazaraa et al. 2006). For the objective function $Z_G = \sum_{i=1}^n D_i P_i^D(T_i, G_i)$, where $D_i \geq 0$, $i = 1, 2, \dots, n$, its Hessian matrix is

$$H_G = \begin{bmatrix} \frac{D_1 \partial^2 P_1(T_1, G_1)}{\partial G_1^2} & \frac{D_1 \partial^2 P_1(T_1, G_1)}{\partial G_1 \partial G_2} & \dots & \frac{D_1 \partial^2 P_1(T_1, G_1)}{\partial G_1 \partial G_n} \\ \frac{D_2 \partial^2 P_2(T_2, G_2)}{\partial G_2 \partial G_1} & \frac{D_2 \partial^2 P_2(T_2, G_2)}{\partial G_2^2} & \dots & \frac{D_2 \partial^2 P_2(T_2, G_2)}{\partial G_2 \partial G_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{D_n \partial^2 P_n(T_n, G_n)}{\partial G_n \partial G_1} & \frac{D_n \partial^2 P_n(T_n, G_n)}{\partial G_n \partial G_2} & \dots & \frac{D_n \partial^2 P_n(T_n, G_n)}{\partial G_n^2} \end{bmatrix}.$$

Based on the assumption in Equation (1), and thus we know that $\partial^2 Z_G / \partial G_i^2 = D_i \partial^2 P_i(T_i, G_i) / \partial G_i^2 \geq 0$ and $D_i \geq 0$. Also, we have $\partial^2 Z_G / \partial G_i \partial G_j = D_i \partial^2 P_i(T_i, G_i) / \partial G_i \partial G_j = 0$, where $i \neq j$, $i, j = 1, 2, \dots, n$. Then we have that the determinant of the Hessian matrix is

$$|H_G| = \begin{vmatrix} \frac{\partial^2 Z_G}{\partial G_1^2} & 0 & \dots & 0 \\ 0 & \frac{\partial^2 Z_G}{\partial G_2^2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{\partial^2 Z_G}{\partial G_n^2} \end{vmatrix} = \prod_{i=1}^n \frac{\partial^2 Z_G}{\partial G_i^2} \geq 0.$$

Thus, H_G is positive semidefinite. So $Z_G(T, G)$ is a concave function (Bazaraa et al. 2006).

For the constraints in problem (4)–(6), since the constraint of the government minimization model $\sum_{i=1}^n G_i = \mathcal{G}$ is an affine (linear) function, then the feasible set $G = \{G_i: \sum_{i=1}^n G_i - \mathcal{G} = 0\}$ is convex (Bazaraa et al. 2006). Thus, the defender's optimization model is a convex problem.

Second, since there are nonnegative constraints in models (4)–(6), to solve the nonlinear problems in (4)–(6) by applying the appropriate KKT conditions, all the nonnegative constraints $G_i \geq 0$ are rewritten as $-G_i \leq 0$, $\forall i = 1, 2, \dots, n$. These nonnegative constraints would get Lagrange multipliers of their own.

From the assumption in Section 2.2, we note that the probability function $P_i(T_i, G_i)$ of a successful attack is continuous and differentiable. Because D_i is a constant, the objective function of the defender $Z_G(T, G) = \sum_{i=1}^n D_i P_i(T_i, G_i)$ is also continuous and differentiable.

Thus, The Lagrangian of the defender's optimization model (4) can be written as follows:

$$L(G, \lambda_G, \mu_G) = - \sum_{i=1}^n D_i P_i(T_i, G_i) + \lambda_G \left(\mathcal{G} - \sum_{i=1}^n G_i \right) + \sum_{i=1}^n \mu_{G_i} (-G_i). \quad (\text{A.1})$$

Note that the budget constraint function $(\mathcal{G} - \sum_{i=1}^n G_i): \mathbb{R}^n \rightarrow \mathbb{R}$ and nonnegative constraints $-G_i$ are continuously differentiable. According to the KKT necessary conditions (Karush 2014), if $\hat{G} = (\hat{G}_1, \dots, \hat{G}_n)$ is a local minimum, there exist some constants (KKT multipliers) μ_{G_i} ($i = 1, \dots, n$) and λ_G such that the following conditions are satisfied:

- Stationarity condition:

$$\Delta L(\hat{G}, \lambda_G, \mu_G) = - \sum_{k=1}^n D_k \frac{\partial (P_k(T_k, \hat{G}_k))}{\partial G_i} - \lambda_G - \mu_{G_i} = 0, \quad i = 1, 2, \dots, n. \quad (\text{A.2})$$

- Primal feasibility condition:

$$\sum_{i=1}^n \hat{G}_i = \mathcal{G}; \quad \hat{G}_i \geq 0, \quad i = 1, 2, \dots, n. \quad (\text{A.3})$$

- Complementary slackness condition:

$$-\hat{G}_i \mu_{G_i} = 0, \quad i = 1, 2, \dots, n. \quad (\text{A.4})$$

- Dual feasibility condition:

$$\mu_{G_i} \geq 0, \quad i = 1, 2, \dots, n. \quad (\text{A.5})$$

From Proposition 1, we note that the defender's optimization model in (4)–(6) is a convex problem, which means that the KKT conditions are sufficient. So, if all the conditions in (A.2)–(A.5) are satisfied, then $\hat{G}$ is a global optimal solution.

A.2. Proof of Proposition 2

To prove that the maximization problem in model (9)–(11) is a convex problem, we need to show that its objective function is concave and the constraints are convex (Bazaraa et al. 2006). Now, let's consider the Hessian matrix of the objective function $Z_T(T, G)$, which is defined as follows:

$$H_T = \begin{bmatrix} \frac{\partial^2 Z_T}{\partial T_1^2} & \frac{\partial^2 Z_T}{\partial T_1 \partial T_2} & \cdots & \frac{\partial^2 Z_T}{\partial T_1 \partial T_n} \\ \frac{\partial^2 Z_T}{\partial T_2 \partial T_1} & \frac{\partial^2 Z_T}{\partial T_2^2} & \cdots & \frac{\partial^2 Z_T}{\partial T_2 \partial T_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 Z_T}{\partial T_n \partial T_1} & \frac{\partial^2 Z_T}{\partial T_n \partial T_2} & \cdots & \frac{\partial^2 Z_T}{\partial T_n^2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{B_1 \partial^2 P_i(T_1, G_1)}{\partial T_1^2} & \frac{B_1 \partial^2 P_i(T_1, G_1)}{\partial T_1 \partial T_2} & \cdots & \frac{B_1 \partial^2 P_i(T_1, G_1)}{\partial T_1 \partial T_n} \\ \frac{B_2 \partial^2 P_i(T_2, G_2)}{\partial T_2 \partial T_1} & \frac{B_2 \partial^2 P_i(T_2, G_2)}{\partial T_2^2} & \cdots & \frac{B_2 \partial^2 P_i(T_2, G_2)}{\partial T_2 \partial T_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{B_n \partial^2 P_i(T_n, G_n)}{\partial T_n \partial T_1} & \frac{B_n \partial^2 P_i(T_n, G_n)}{\partial T_n \partial T_2} & \cdots & \frac{B_n \partial^2 P_i(T_n, G_n)}{\partial T_n^2} \end{bmatrix}.$$

Since $B_i \partial^2 P_i(T_i, G_i) / \partial T_i \partial T_j = 0$, where $i \neq j, i, j = 1, 2, \dots, n$, we have

$$H_T = \begin{bmatrix} \frac{B_1 \partial^2 P_i(T_1, G_1)}{\partial T_1^2} & 0 & \cdots & 0 \\ 0 & \frac{B_2 \partial^2 P_i(T_2, G_2)}{\partial T_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{B_n \partial^2 P_i(T_n, G_n)}{\partial T_n^2} \end{bmatrix}.$$

To study the H_T , we consider the Hessian matrix of $-Z_T(T, G)$:

$$\hat{H}_T = \begin{bmatrix} -\frac{B_1 \partial^2 P_i(T_1, G_1)}{\partial T_1^2} & 0 & \cdots & 0 \\ 0 & -\frac{B_2 \partial^2 P_i(T_2, G_2)}{\partial T_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -\frac{B_n \partial^2 P_i(T_n, G_n)}{\partial T_n^2} \end{bmatrix}.$$

Based on the assumption in (1), we note that $\partial^2 P_i(T_i, G_i) / \partial T_i^2 \leq 0$ and $B_i \geq 0$, and thus we have $-B_i \partial^2 P_i(T_i, G_i) / \partial T_i^2 \geq 0$. Since $\hat{H}_T$ is a diagonal matrix, all the

diagonal elements are nonnegative, and the determinate of $\hat{H}_T$ is

$$|\hat{H}_T| = \begin{vmatrix} -\frac{\partial^2 Z_T}{\partial T_1^2} & 0 & \cdots & 0 \\ 0 & -\frac{\partial^2 Z_T}{\partial T_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -\frac{\partial^2 Z_T}{\partial T_n^2} \end{vmatrix} = \prod_{i=1}^n -\frac{\partial^2 Z_T}{\partial T_i^2} \geq 0.$$

Thus, $\hat{H}_T$ is positive semidefinite. So, we note that H_T is negative semidefinite. We conclude that $Z_T(T, G)$ is a concave function. Now, we know that the objective function of the terrorist's maximization model (9) is concave. The equality constraint in (9) is an affine function, which is also a convex set. Thus, we know that terrorist's maximization model is a convex problem.

The proof of the second part of Proposition 2 is similar to the proof of Proposition 1 and thus omitted for space considerations.

A.3. Proof of Proposition 3

According to the $\hat{T}(G)$ defined in Proposition 2, we add all the KKT conditions in Equation (13) as the constraints to the defender's optimization problem in Equations (16)–(18). Then the defender's optimization problem can be rewritten as follows:

$$\min_G Z_G = \sum_{i=1}^n D_i P_i(\hat{T}_i, G_i) \quad (\text{A.6})$$

$$\text{subject to: } \sum_{i=1}^n G_i = \mathcal{G}, \quad (\text{A.7})$$

$$G_i \geq 0, \quad i = 1, 2, \dots, n, \quad (\text{A.8})$$

$$\sum_{k=1}^n B_k \frac{\partial (P_k(\hat{T}_k, G_k))}{\partial T_i} - \lambda_T - \mu_{T_i} = 0, \quad \forall i = 1, 2, \dots, n, \quad (\text{A.9})$$

$$\sum_{i=1}^n \hat{T}_i = \mathcal{T}, \quad (\text{A.10})$$

$$\hat{T}_i \geq 0, \quad \forall i = 1, 2, \dots, n, \quad (\text{A.11})$$

$$\bar{\mu}_{T_i} \hat{T}_i = 0, \quad \forall i = 1, 2, \dots, n, \quad (\text{A.12})$$

$$\bar{\mu}_{T_i} \geq 0, \quad \forall i = 1, 2, \dots, n. \quad (\text{A.13})$$

Since all the necessary conditions in Equations (16)–(18) are sufficient, an optimal solution (T^*, G^*) solved by applying equation system (21) specified in Proposition 3 is an SPNE as defined in Definition 2 in Section 3.2.

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